Variant Interpretation in CYP2C19

BIOT 6900 · Module 5 Assignment (Computational Lab 5) · Fall 2026 · Sharath Mandula

Gene: CYP2C19 · Predictor calibrated: CADD (pooled with DPYD) · AlphaGenome Atlas: not run (disclosed below)

Summary. My fetch returned 80 ClinVar variants in CYP2C19. Both anchor alleles came back: CYP2C19*2 (rs4244285, gnomAD AF 0.168) and CYP2C19*17 (rs12248560, AF 0.200). CADD, REVEL, AlphaMissense and gnomAD were populated for 85.0%, 67.5%, 67.5% and 68.8% of rows. The striking result is the label distribution: 38 ‘other’ (almost all drug response), 33 VUS, 9 B/LB and **zero P/LP**. CYP2C19 sits entirely off the pathogenic–benign axis, and almost every result below follows from that.

6.1 · Thresholds and calibration

Threshold. I set MAX_AF = 0.30. The commonest functional alleles in this gene are *17 (global AF 0.200 in my data) and *2 (0.168, and roughly 30% in East Asian populations). The threshold sits above both with a margin. The biological reason a frequency filter has to be loose here is that a CYP2C19 loss-of-function allele carries no fitness cost. A *2/*2 poor metaboliser is healthy and only becomes a clinical problem when given clopidogrel, a drug that has existed for about 25 years. Purifying selection has never acted on these alleles, so their frequency says nothing about their function. In a rare-disease gene, a variant at 15% cannot cause a severe penetrant disorder. In CYP2C19, the most important variant in the gene is at 15%. At 0.30 the filter removed only 2 of 80 variants, and neither was P/LP, drug response or an anchor. A filter that barely acts is the correct result for this gene. One weakness: gnomad_af in this pipeline is the *global* frequency. A popmax-based threshold would be more defensible, because global AF dilutes population-specific highs.

The 0.001 contrast. A rare-disease filter would have removed 19 variants, 15 of them drug-response or anchor variants. These include *2 (rs4244285, AF 0.168) and *17 (rs12248560, 0.200), the deep intronic rs12769205 (0.173), and rs12777823 (0.190). It also removes the rarer no-function alleles *3 (rs4986893, W212X, 0.0025), *4 (rs28399504, M1V start-lost, 0.0024) and *8 (rs41291556, W120R, 0.0016). Most of the alleles that CPIC uses to assign metaboliser phenotype for clopidogrel would therefore have been deleted before any predictor saw them.

Calibration route (disclosed fallback). I calibrated CADD because it is the only tool that scores every consequence class (85.0% overall, versus 67.5% for REVEL, which is missense-only). CYP2C19 alone could not be calibrated. It had **0 P/LP and 5 B/LB** variants with a CADD score, and no likelihood ratio can be computed with an empty numerator. I took the pooling route and added DPYD (426 variants), giving **111 labelled variants (39 P/LP, 72 B/LB)**.

CADD band	n	P/LP	B/LB	LR+ (raw)	LR+ (0.5 correction*)	Implied strength	Reliable?
(0,10]	47	0	47	0.00	0.019	strong, benign	zero P/LP cell
(10,20]	19	0	19	0.00	0.047	strong, benign	zero P/LP cell
(20,25]	10	5	5	1.85	1.85	no evidence either way	borderline (n=10)
(25,30]	4	3	1	5.54	4.31	moderate, pathogenic	NO (n=4)
(30,100]	31	31	0	inf	116	strong, pathogenic	zero B/LB cell

**Sensitivity check I added by hand: 0.5 added to each cell count, so the zero cells give a finite estimate instead of 0 or infinity.*

What this does to my confidence. Very little of this table is evidence *about CYP2C19*. Every one of the 39 P/LP variants comes from DPYD, where P/LP means DPD deficiency, a Mendelian loss-of-function disorder that does sit on the ACMG axis. The (25,30] band rests on 4 variants and I did not apply it. Three bands have an empty cell, so their raw LRs of 0 and infinity are artefacts of perfect separation in a small sample, not measurements. The decisive check is the gene's own anchor. **CYP2C19*2 has CADD 5.686, which places it in the (0,10] band:**

‘strong, benign’. The calibration would assign strong benign evidence to the best-established no-function allele in the gene. *17 (CADD 3.031) lands in the same band. Its ClinVar string ‘Benign; other’ made clinvar_label() count an increased-function allele as one of the 5 CYP2C19 B/LB training points. So the calibration is internally consistent but transfers badly. In Q5 it moved 16 of 33 VUS ‘towards benign (strong)’, 0 towards pathogenic, and left 17 in place (8 in the LR~1 band, 6 with no CADD score, 3 in the n=4 band). I would not act on those 16 benign moves. The most I would allow them in this gene is supporting-level BP4, and only for variants without a splicing or regulatory hypothesis.

6.2 · Coverage gaps and the regulatory genome

REVEL and AlphaMissense together scored nothing in eight classes: synonymous (n=7), upstream (5), intron (1), intergenic (1), splice donor (1), stop-lost (1), stop-gained (1) and 8 rows with no snpEff annotation at all. That is 25 variants a missense-only pipeline would skip without warning. CADD filled most of this gap, but not all of it. The single splice_donor_variant and all 8 unannotated rows had no CADD score either, and the discordance cell flagged 10 variants that no tool could score. Those blanks mean ‘could not score’, not ‘benign’. The most important gap is also the subtlest: *2 is annotated **synonymous**. Its real mechanism is splicing. c.681G>A creates an aberrant splice site in exon 5 that removes 40 bp and leads to a premature stop. The collapsed consequence term, the missense tools and the CADD calibration all miss it.

My unscored variant: CYP2C19*17, rs12248560 (c.-806C>T). It is upstream, so REVEL and AlphaMissense are blank, and its CADD of 3.031 is low because the promoter position is weakly constrained. To classify it I would need evidence that the variant changes transcription in the tissue where the drug is activated, which is the liver. Two sources would give that. A **luciferase reporter assay** in HepG2 cells, comparing C and T promoter constructs, would test whether the T allele increases expression, as expected if it creates a transcription-factor (GATA) binding site. **Liver eQTL data** (GTEx) showing allele-specific CYP2C19 expression would test the same thing in tissue. This is evidence of an increased-function effect, which the pathogenic/benign framework cannot represent at all.

The Atlas (not run). The alphagenome package was not installed in my environment, so the Atlas cell raised an ImportError and reported ‘not run’. No AVI scores were used anywhere in this analysis. In principle the Atlas would have been the only tool in the course able to score my 25 non-missense variants by mechanism rather than by constraint. I would expect *17 and rs7902257 (upstream, AF 0.022, drug response, no score from any tool) to show regulatory attributions (MAX_ABS_DNASE / CHIP_TF in liver), and *2 and the splice_donor_variant to show MERGED_SPLICING. If either top score were instead driven by CACTUS_241_WAY, that would tell me conservation was carrying it and the mechanism was still unresolved. That outcome is plausible in a gene under weak selection. Nothing else in the course scores regulatory variants by mechanism: CADD's low scores for *17 and *2 reflect constraint, not function.

6.3 · A discordant variant, read mechanistically

rs41291556, CYP2C19*8 (W120R). It tripped **revel_vs_am**. REVEL scored it 0.440, AlphaMissense 0.940 (a gap of 0.50), and CADD 25.7. Its gnomAD AF is 0.0016 and ClinVar files it as ‘other’, not P/LP. CPIC classifies *8 as a no-function allele.

For both outputs to make sense, the following would have to be true: the substitution genuinely destroys the enzyme, but it does so in a gene whose variants are never disease-causing. I think that is what is happening. AlphaMissense is label-free. It reads evolutionary and structural plausibility, and replacing a buried, bulky aromatic tryptophan with a charged arginine is exactly the kind of change it flags as destabilising. That matches *8's no-function status. REVEL is an ensemble trained on disease labels, pathogenic versus neutral missense. A no-function CYP2C19 allele causes no disease, and its component conservation scores are diluted in a gene that has paralogs across the CYP2C cluster and is under weak purifying selection. So REVEL is not wrong about the molecule. It is answering ‘does this look like a Mendelian disease variant?’, and the honest answer to that is

‘moderately’. I prefer this explanation to the alternative that the ClinVar label is unreliable. The functional evidence for *8 is consistent, and the same pattern repeats in this gene: R186P (REVEL 0.209, AM 0.880) and F448L (REVEL 0.254, AM 0.798) show the same split. The discordance is systematic, not noise. For a pharmacogene, the label-free structural model is the more appropriate instrument.

6.4 · What I could not determine

My most interesting variant is **CYP2C19*2 (rs4244285)**. It is the most consequential allele in the gene, my pipeline labelled it synonymous, and my calibration would have scored it ‘strong, benign’. Two things my analysis cannot establish about it:

1. Zygosity and phase. Clopidogrel response depends on diplotype. A *1/*2 intermediate metaboliser and a *2/*2 poor metaboliser receive different recommendations, and *2 in combination with *17 depends on which haplotypes they sit on. ClinVar records single variants with no genotype, so nothing here tells me how many copies a patient carries or how *2 is phased with *17 or with the linked intronic rs12769205.

2. Modifiers, including phenoconversion. Even a correctly genotyped *1/*2 patient can behave as a poor metaboliser when taking a CYP2C19 inhibitor such as omeprazole, a proton-pump inhibitor often co-prescribed with clopidogrel. Predicted phenotype from genotype is not observed phenotype, and none of my data sees co-medication.

The single experiment. An RNA assay would test the call my pipeline actually got wrong: RT-PCR (or targeted RNA-seq) of CYP2C19 transcripts from primary human hepatocytes heterozygous for rs4244285, measuring the fraction of transcripts that skip the first 40 bp of exon 5. A substantial aberrant isoform from the A allele, with little normal product, would confirm a splicing loss-of-function mechanism and show that the CADD-based ‘strong, benign’ band is wrong for this variant class. **What would change my mind:** if the A allele produced normally spliced transcript at levels similar to the G allele, the splice explanation would fail. I would then have to look for another mechanism, such as linkage to a causal variant like rs12769205, before accepting *2 as functional.

Fallbacks disclosed

(1) CYP2C19 had 0 P/LP labelled variants, so CADD was calibrated on a pool with DPYD. Every P/LP variant comes from DPYD, so the bands describe DPYD-like loss of function, not CYP2C19 pharmacology. (2) The AlphaGenome Atlas was not run because the client was not installed. No AVI scores were used, and the Atlas discussion above is reasoned, not measured. (3) Allele frequency is global, not popmax. (4) The 0.5-correction LRs in the 6.1 table were computed by hand as a sensitivity check. They are not notebook output.